

Roll No.:47

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Write a python code to demonstrate importing pandas libraries and create data frame object.

```
import pandas as pd

data = {
    'Name': ['Komal', 'Laxmi', 'Anjali'],
    'Marks': [85, 90, 78],
    'City': ['Mumbai', 'Pune', 'Nashik']
}

df = pd.DataFrame(data)

print(df)
```

	Name	Marks	City
0	Komal	85	Mumbai
1	Laxmi	90	Pune
2	Anjali	78	Nashik

Write a python code to show statistical information on given data set.

```
# Import statistics Library
import statistics

# Calculate average values
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
```

```
7
6
1.8666666666666667
```

```
# Import statistics Library
import statistics

# Calculate middle values
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
```

```
7
6.0
```

1.05

```
# Import statistics Library
import statistics

# Calculate the mode
print(statistics.mode([1, 3, 3, 3, 5, 7, 7, 9, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
```

```
3
1
red
3
1
red
```

```
# Import statistics Library
import statistics

# Calculate the variance from a sample of data
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
```

```
14
0.47966666666666663
70.80333333333333
1736.9166666666667
```

.Write a python code to create pandas series from dictionaries.

```
import pandas as pd

data = {
    'Komal': 85,
    'Laxmi': 90,
    'Anjali': 78
}

series = pd.Series(data)

print(series)
```

```
Komal    85
Laxmi    90
Anjali    78
dtype: int64
```

Write a python code to demonstrate filter pandas series with Boolean arrays.

```
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["jyoti", "sachin", "parul", "laxmi"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
```

	name	degree	score
True	jyoti	MBA	90
False	sachin	BCA	40
True	parul	M.Tech	80
False	laxmi	MBA	98